TOTAL QUALITY MANAGEMENT

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TQM

- **Definition:** Total Quality Management is a systematic approach aimed at enhancing the quality of products and services by involving all levels of an organization.

- **Key Focus:** Customer satisfaction, continuous improvement, teamwork.

- **Importance:** TQM integrates quality into every phase of the business process.

Axioms of TQM

- Commitment (to never ending quality improvement & innovation).
- Scientific knowledge (of the proper tools and techniques for the ‘technical change’).
- Involvement (All in one team for the social change)

Dr. William Edwards Deming (October 14, 1900 – December 20, 1993)

- Engineer, statistician, professor, author, lecturer, and management consultant
- Deming was asked to go to Japan after World War II to help with the Japanese census, and later to improve radio manufacturing.
- He made a significant contribution to Japan's reputation for innovative, high-quality products after World War II.
- He was honored in Japan in 1951 with the establishment of the Deming Prize.

Introduction to Deming's Philosophy

- W. Edwards Deming: Known as the father of quality management. Deming’s principles revolutionized management thinking, particularly in Japan post-WWII.
- **Key Contribution:** His focus on **statistical control** and **continuous improvement** shaped modern TQM practices.
- Inspired by: Walter Shewhart, who provided the concept of statistical process control in 1924.

Deming's 14 Points for Management (Slide 1)

1. Create constancy of purpose for improving products and services.
2. Adopt the new philosophy.
3. Cease dependence on inspection to achieve quality.
4. End the practice of awarding business on the price tag alone.
5. Improve constantly and forever every process for planning, production, and service.
6. Institute training on the job.
7. Adopt and institute leadership.

Deming's 14 Points for Management (Slide 2)

8. Drive out fear.
9. Break down barriers between staff areas.
10. Eliminate slogans, exhortations, and targets for the workforce.
11. Eliminate numerical quotas for the workforce and management.
12. Remove barriers to pride of workmanship.
13. Encourage education and self-improvement for everyone.
14. Take action to accomplish the transformation.

Importance of Intrepidness in Gita

अभयं सत्त्वसंशुद्धिः ज्ञानयोगव्यवस्थितिः।

दानं दमश्च यज्ञश्च स्वाध्यायस्तप आर्जवम्।।16.1।।

अहिंसा सत्यमक्रोधस्- त्यागः शान्तिरपैशुनम् ।

दया भूतेष्वलोलुप्त्वं मार्दवं ह्रीरचापलम् ॥16.2॥

तेजः क्षमा धृतिः शौचमद्रोहो नातिमानिता ।

भवन्ति संपदं दैवीम-भिजातस्य भारत ॥16.3॥

Deming's System of Profound Knowledge

- **Appreciation for a System:** Businesses are interconnected systems that require all parts to work together.

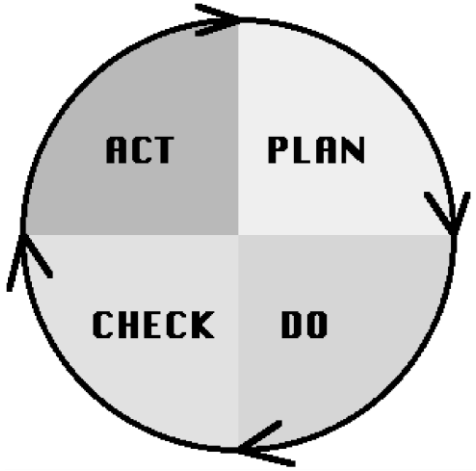
- **Knowledge of Variation:** Understanding variation is crucial to improving performance.

- **Theory of Knowledge:** Decisions must be grounded in theory and data, not just assumptions.

- **Psychology:** Understanding human motivation is essential for effective management.

Plan-Do-Check-Act (PDCA) Cycle

- **Plan:** Identify a goal or purpose, formulate a theory, define success metrics, and put a plan into action.
- **Do:** Implement the plan on a small scale.
- **Check:** Measure and analyze the results.
- **Act:** Take action based on what was learned in the study phase.



Deming's Focus on Continuous Improvement

- **Kaizen:** Japanese concept emphasizing continuous, incremental improvement.

- **Cycle of Improvement:** Deming's philosophy encourages constant reflection on processes and finding ways to improve.

Seven Deadly Sins

- Lack of constancy
- Short-term profits
- Performance appraisal
- Job-hopping
- Use of visible figures only
- Excessive medical costs
- Excessive costs of liability

Deming’s "Chain Reaction" Theory

Improving quality leads to reduced costs, which increases productivity, resulting in capturing markets and sustaining long-term business success.

Comparison of TQM Philosophies: Juran's Trilogy

Joseph Moses Juran (1904—2008), Romanian-born American engineer, management consultant and author

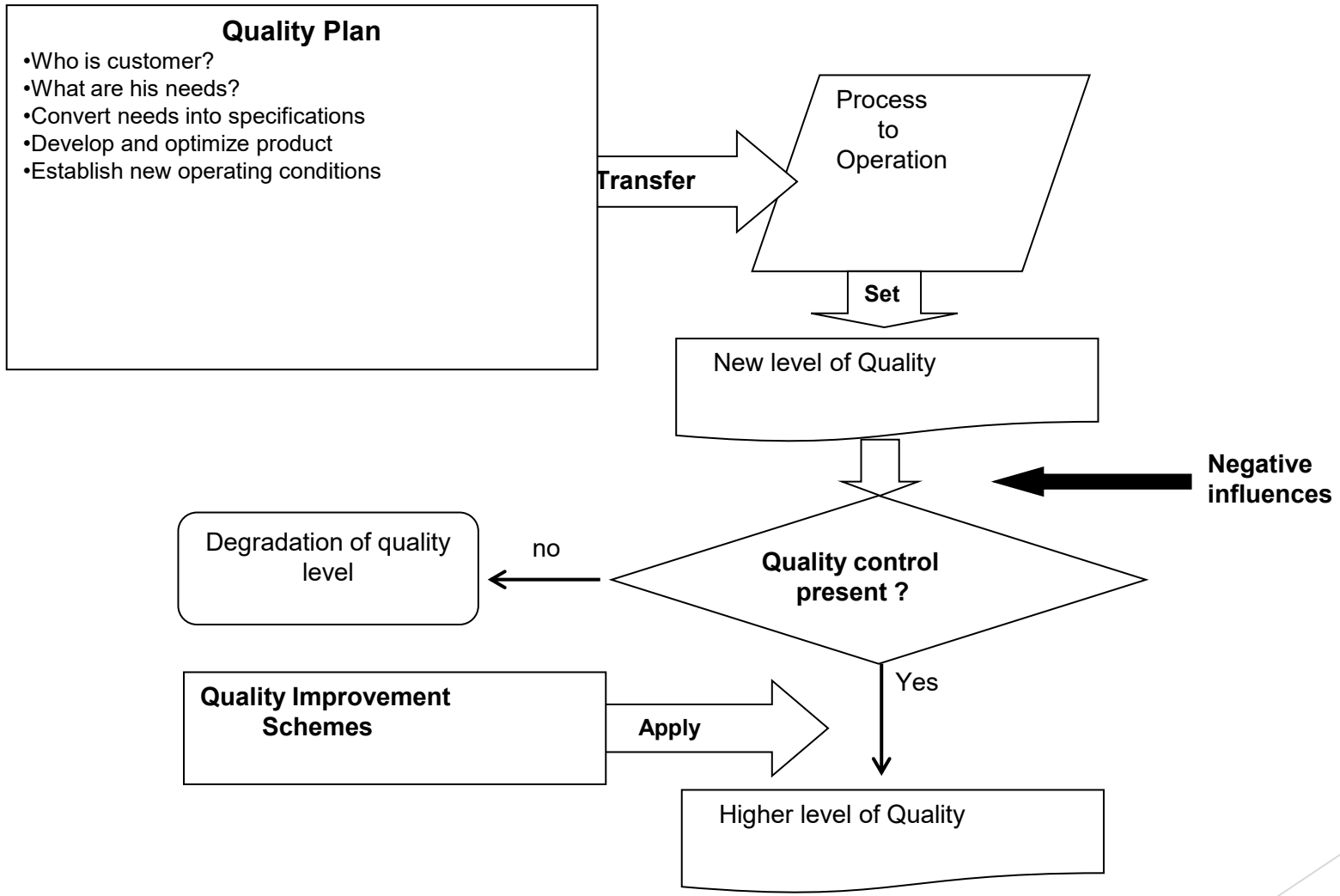
- **Juran's Key Idea:** Fitness for use and emphasis on managing quality as part of strategic planning.

- **Difference from Deming:** Juran emphasizes the managerial aspects of quality, whereas Deming focuses on systemic improvements.

- **Juran's Philosophy:** Focus on planning, control, and improvement of quality.

- Management for quality is done by the use of three managerial processes- *planning*, *control* and *improvement* which are three members of Juran's Quality Trilogy model.

Quality Trilogy Model



Comparison of TQM Philosophies: Philip B. Crosby (1926—2001)

Crosby initiated the Zero Defects program at the Martin Company.

- **Crosby's Motto:** "Quality is Free" – investing in quality reduces costs in the long run.

- **Zero Defects:** Belief that quality should be built into processes, and companies should aim for **zero defects**.

- **Difference from Deming:** Crosby takes a more **absolutist approach** with zero tolerance for errors, whereas Deming focuses on reducing variation.

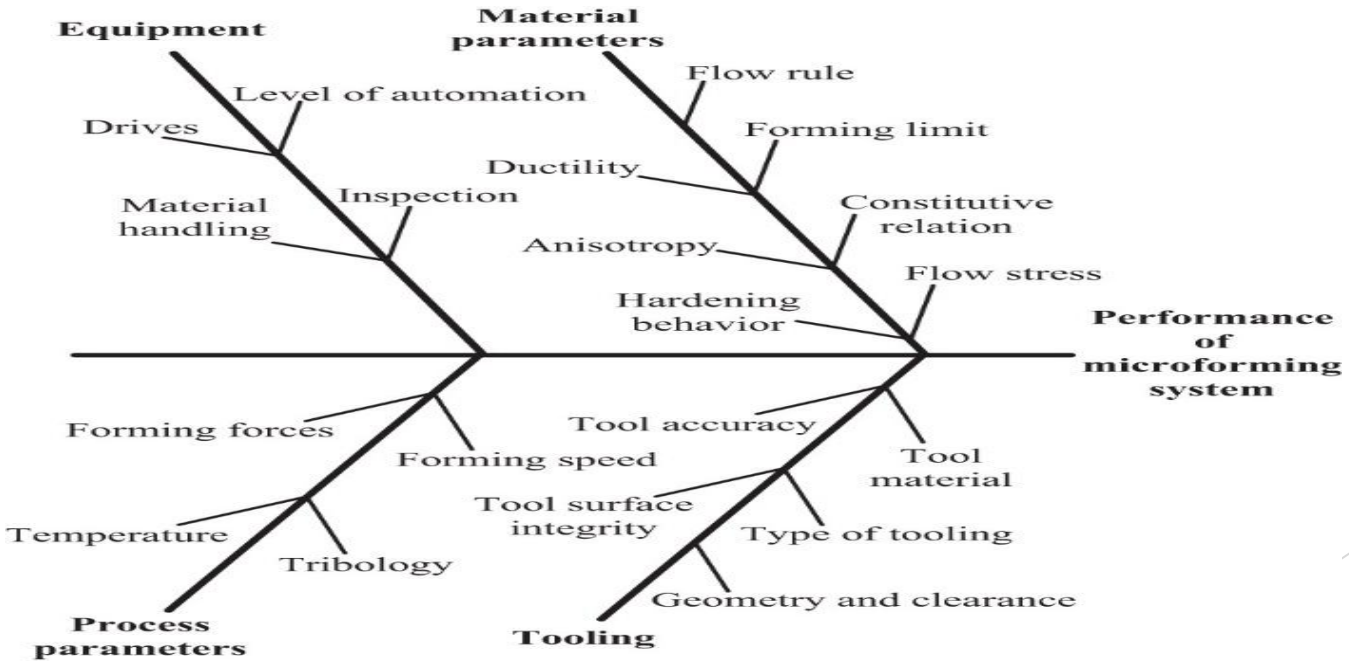
Comparison of TQM Philosophies: Armand V. Feigenbaum (1920—2014)

- **Total Quality Control (TQC):** Focuses on the organization as a whole, incorporating every department in the quality process.

- **Difference from Deming:** Feigenbaum takes a broader approach, incorporating more organizational levels into the quality effort, compared to Deming's focus on systemic processes.

Comparison of TQM Philosophies: Kaoru Ishikawa (1915—1989)

- **Quality Circles:** A bottom-up approach to involve workers in quality discussions.
- **Fishbone Diagram (Ishikawa Diagram):** A visual method for identifying cause-and-effect relationships in quality issues.
- **Difference from Deming:** Ishikawa focuses heavily on employee involvement at all levels.



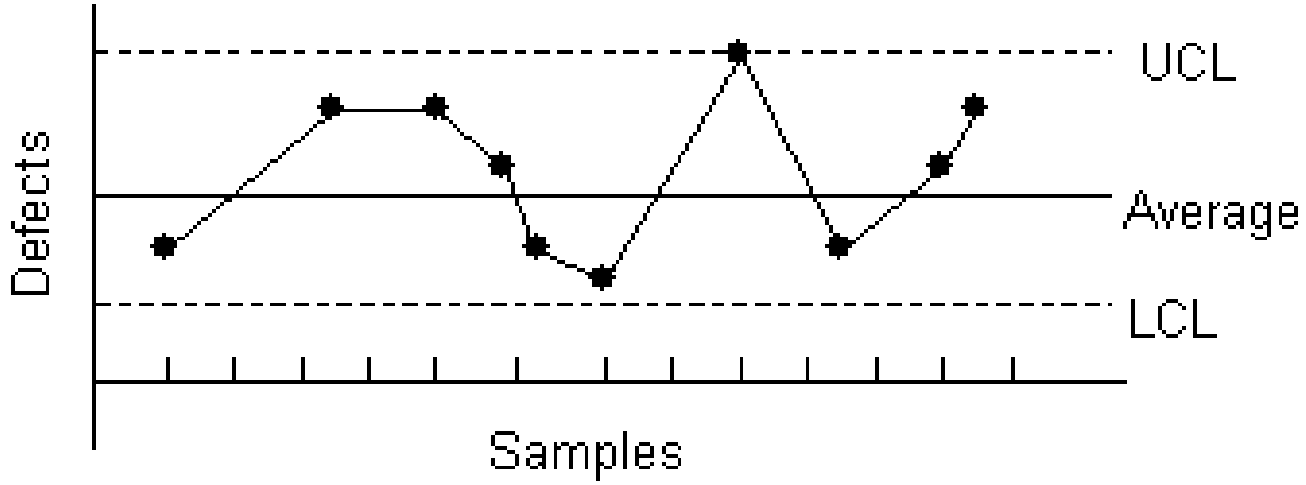
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Comparison of TQM Philosophies: Genichi Taguchi (1924—2012)

- **Loss Function:** Taguchi defined quality as the **minimization of societal loss** caused by variation in products.
- **Design of Experiments (DOE):** A robust methodology for improving product and process quality.
- **Difference from Deming:** Taguchi's focus is on design and production stages, while Deming emphasizes ongoing, system-wide improvements.

The Role of Statistical Process Control (SPC)

- **SPC:** Statistical methods to monitor and control processes.
- Deming pioneered the use of SPC to ensure consistency and improvement.
- **Key Tools:** Control charts, variance analysis



Famous quotes by Deming

- **A company could put a top man at every position and be swallowed by a competitor with people only half as good, but who are working together.**
- Relevant Indian view:

संगच्छध्वं संवदध्वं सं वो मनांसि जानताम्। देवा भागं यथा पूर्वे सञ्जानाना उपासते।।

- We installed quality control.' No. You can install a new desk, or a new carpet, or a new dean, but not quality control. Anyone that proposes to 'install quality control' unfortunately has little knowledge about quality control.
- A leader is coach and counsel, not a judge
- A dissatisfied customer does not complain: he just switches.
- Good quality and the right uniformity have no meaning except with reference to the consumer's demands.
- A bad system will beat a good person every time.

Some interesting quotes by Deming

- No matter how good your people are, there will always be someone at the bottom, even if the differences are insignificant.
- The greatest waste in America is failure to use the abilities of people.
- The performance of anybody is governed largely by the system he works in.
- Evaluation of performance, merit rating, or annual review... destroys people. It destroys teamwork.
- A bad system will beat a good person every time.

मुखिया मुख सों चाहिए, खान पान को एक।
पालै पोसै सकल अंग, तुलसी सहित विवेक॥

Some more quotes (Slide 1)

- The United States is the most underdeveloped nation in the world in terms of managing people
- Someone told me that there were five candidates for president of one of the biggest and most famous of America's companies. When one of them was selected, other four resigned from the company. Such a thing could not happen in Japan. So you see, the American system is so set up that it cannot retain the talents of its people. That's very serious.
- People have to feel secure. That means, 'without concern' from the Latin 'se' for 'without' and 'cure' meaning 'care' or 'concern'.

Some more quotes (Slide 2)

- Quality means making it right the first time, so that you do not have to rework.
- If you can't describe what you are doing as a process, you don't know what you're doing
- It's not enough to do your best; you must know what to do, and then do your best.
- You do not have to do it; survival is not compulsory.
- Managers who visit other companies and try to learn how things are done are wasting their time. They will come back with a few anecdotes and feel good about their trip, but they will not improve their own organization

Genichi Taguchi won the Deming Prize twice. He first won it in 1987 for his contributions to quality control and his development of robust design methodology. He won it again in 1991 for his continued work and advancements in the field of quality management. The Deming Prize is a prestigious award given for contributions to the field of quality management, and Taguchi's work significantly influenced the field.

Case Study 1: Toyota – Applying Deming's Quality Management

- **Overview:**
  - Toyota adopted Deming's principles in the 1950s, leading to the development of the **Toyota Production System (TPS)**.
  - Emphasized **continuous improvement (Kaizen)**, waste reduction, and efficiency.
- **Deming's Influence:**
  - Statistical Process Control (SPC) was used to minimize variability.
  - Empowered employees at all levels to contribute to quality improvements.
- **Outcome:**
  - Toyota became a global leader in automotive quality, known for reliability and efficiency.
- **Key Lesson:** Systemic thinking and employee involvement can transform production processes.

Slide 2: Case Study 2: Wipro – Blending Deming's Philosophy with Indian Management

•Overview:

- Wipro, an Indian multinational IT services company, applied both **Deming's TQM** and Indian philosophical principles like **Karma** and **Dharma** in management.

•Application:

- Implemented **Six Sigma**, focusing on process improvement and quality control, inspired by Deming.
- Integrated **Dharma (Duty)** into organizational culture, encouraging ethical behavior and social responsibility.

•Outcome:

- Wipro achieved high operational excellence and became a global leader in IT services.

•Key Lesson: Combining process improvement with ethical frameworks strengthens both performance and culture.

Case Study 3: Motorola – Deming's 14 Points in Action

- **Overview:**
  - In the 1980s, Motorola used **Deming's 14 Points for Management** to develop the **Six Sigma methodology**.
- **Deming's Influence:**
  - Focused on **eliminating defects** and improving quality across all levels of production.
  - Emphasized continuous training, leadership, and a commitment to long-term quality.
- **Outcome:**
  - Motorola drastically reduced errors in manufacturing and won the **Malcolm Baldrige National Quality Award**.
- **Key Lesson:** Long-term commitment to quality and leadership involvement can lead to global recognition.

Case Study 4: Tata Group – Indian Philosophy in Business

•**Overview:**

- Tata Group integrates **Indian philosophical values** like **Ahimsa (non-violence)** and **Sustainability** in its business practices.

•**Application:**

- Promotes **inclusive growth**, focusing on social impact and ethical business practices.
- Emphasizes **Ahimsa** through corporate responsibility, ensuring no harm to communities or the environment.

•**Outcome:**

- Tata Group has become a leader in **corporate social responsibility (CSR)**, with a strong global reputation for ethical business.

•**Key Lesson:** Incorporating Indian philosophical principles into business leads to sustained ethical growth and social impact.

Case Study 5: Ford Motor Company – Deming's Transformation in the 1980s

- **Overview:**
  - Ford faced declining quality and market share in the 1980s, leading to Deming being invited to transform its quality management.
- **Deming's Influence:**
  - Introduced **statistical process control** and eliminated unnecessary inspections.
  - Implemented the concept of **"Quality is everyone's responsibility."**
- **Outcome:**
  - Ford significantly improved quality, leading to the success of models like the **Ford Taurus**, and regained market share.
- **Key Lesson:** Systemic change and leadership commitment can revive failing companies.

Case Study 6: Infosys – Integration of Deming's and Indian Ethical Principles

- **Overview:**
  - Infosys, an Indian multinational corporation, used **Deming's TQM** along with Indian ethics like **Satyam (truth)** and **Seva (service)**.
- **Application:**
  - Focused on **client satisfaction** and **continuous process improvement** using Deming's principles.
  - Embedded **Indian ethical values** into its leadership and corporate governance, fostering transparency and responsibility.
- **Outcome:**
  - Infosys became known for its **ethical business model**, receiving global recognition for both its quality management and leadership.
- **Key Lesson:** Blending process excellence with ethical leadership creates long-term sustainable success.